WORKSHEET - NAPLAN Year 9

Number and Algebra
Patterns (9)

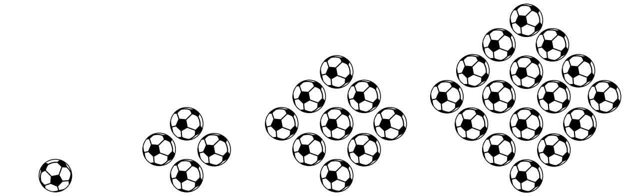
Teacher: Joanne Rust

Exam Equivalent Time: 8.75 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Number and Algebra, NAP-78013

Flynn is making a pattern with soccer balls.



Pattern #1 Pattern #2 Pattern #3 Pattern #4

This table shows the number of balls he needs for each shape in the pattern.

Pattern #	1	2	3	4	5
Number of balls	1	4	9	16	**?**

How many balls will Flynn need for Pattern #5?

17 **23** **25** **30**

2. Number and Algebra, NAP-42660

The table below shows a pattern.

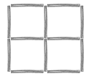
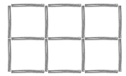
Top number	1	2	3	4	...	?
Bottom number	4	8	12	16	...	28

What is the top number when the bottom number is 28?

5 **6** **7** **9**

3. Number and Algebra, NAP-66185

David makes the following pattern of figures using sticks.

Figure 1	Figure 2	Figure 3		Figure 6
			...	?
7 sticks	12 sticks	17 sticks	...	? sticks

How many sticks will Figure 6 require?

30 **32** **34** **37**

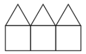
4. Number and Algebra, NAP-19175

Which expression is equal to $6^3 \times 36^2$?

- $6 \times 3 \times 36 \times 2$
- $6 \times 6 \times 6 \times 6 \times 6$
- $6 \times 6 \times 6 \times 36 \times 6$
- $6 \times 6 \times 6 \times 6 \times 6 \times 6 \times 6$

5. Number and Algebra, NAP-95680

Peter uses matchsticks to make a pattern of shapes, as shown in the table below.

Shape number (N)	1 	2 	3 
Number of Sticks (S)	6	11	16

The equation used to show the relationship between S and N is

$S = N + 5$ $S = 5N$ $S = 5N + 1$ $S = 7N - 1$

6. Number and Algebra, NAP-95681

Dan is a lettuce grower and the cost of fertilising different crop areas is shown in the table below.

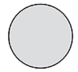
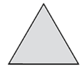
size of crop area (m^2)	30	40	50	60	70
cost of fertilising ($)	2100	2800	3500	4200	4900

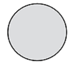
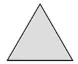
Complete the rule that agrees with the values in the table.

cost of fertilising crop = $\boxed{}$ × *area of crop*

7. Number and Algebra, NAP-13288

Sharon made a number pattern using this rule:

next number = previous number ×  + 

 and  are whole numbers.

Her number pattern is:

$-2, -1, 1, 5, 13, \ldots$

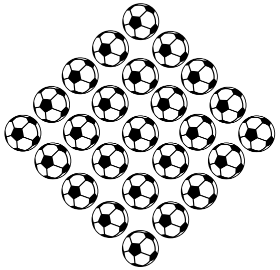
What is the next number in Sharon's pattern?

17 19 21 29

Worked Solutions

1. Number and Algebra, NAP-78013

Pattern #5 is



$\therefore$ 25 balls needed for Pattern #5.

---

2. Number and Algebra, NAP-42660

The pattern is:

Top number $\times$ 4 $=$ Bottom number

$\therefore$ Top number $= \dfrac{28}{4} = 7$

---

3. Number and Algebra, NAP-66185

Each figure adds 5 extra sticks,

$\Rightarrow$ Figure 4 - 22 sticks

$\Rightarrow$ Figure 5 - 27 sticks

$\therefore$ Figure 6 - 32 sticks

---

4. Number and Algebra, NAP-19175

$$\begin{aligned}6^3 \times 36^2 &= (6\times 6\times 6)\times(6\times 6)^2\\ &= (6\times 6\times 6)\times(6\times 6)\times(6\times 6)\\ &= 6\times 6\times 6\times 6\times 6\times 6\times 6\end{aligned}$$

Worked Solutions

5. Number and Algebra, NAP-95680

Shape 1: $S = 5\times 1 + 1 = 6$

Shape 2: $S = 5\times 2 + 1 = 11$

Shape 3: $S = 5\times 3 + 1 = 16$

$\therefore S = 5N + 1$

---

6. Number and Algebra, NAP-95681

$$\begin{aligned}\text{Cost of fertilising} &= \text{cost of 1 m}^2 \times \text{area of crop}\\ &= \frac{2100}{30}\times\text{area of crop}\\ &= 70\times\text{area of crop}\end{aligned}$$

7. Number and Algebra, NAP-13288

Using the rule:

$$-1 = -2 \times \bigcirc + \triangle \qquad \ldots (1)$$

$$1 = -1 \times \bigcirc + \triangle \qquad \ldots (2)$$

Subtract $(1) - (2)$,

$$-2 = -\bigcirc$$

$$\bigcirc = 2$$

Substitute $\bigcirc = 2$ into (1)

$$-1 = -4 + \triangle$$

$$\triangle = 3$$

$$\therefore \text{ Next number} = 13 \times 2 + 3$$
$$= 29$$